Probabilistic Combinatorics

Sheet 8

June 10, 2024

Problem 1. Let again $G = ([n], E)$ an undirected graph and $F_1, \dots, F_n \subseteq F$ a collection of possible colors for every node, i.e. F is a superset of colours and F_i is the set of colours allowed for the vertex i. Here let $\#F_i = 8r$ for every $i \in [n], r > 0$. Further for every $c \in F_i$, let there be at most r neighbours j of i, s.t c is also allowed for j, i.e.

$$\#\{j \in N(i) \mid c \in F_j\} \leq r.$$

Show: There is an allowed colouring f (i.e. No edge is monochromatic) of G, with $f(i) \in F_i$.

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sampling space Ω is the set of all possible vertex colourings f of the graph G where $f(i) \in F_i$ and $i \in [n]$, and the probability space $\mathbb{P}(\omega) = \left(\frac{1}{8r}\right)^{|E|}$ for any vertex colouring $\omega \in \Omega$. For any edge $e \in E$, and any colour $c \in F$ that is allowed to colour both endpoints of e, let $A_{e,c}$ be an event in the probability space $(\Omega, \mathbb{P})$ where the edge e is non-monochromatic, or the two endpoints of the edge e are coloured c. For each endpoint of a edge e, there are $8r$ colours that we can choose to colour it, then the probability of an endpoint of the edge e to be coloured c is $\frac{1}{8r}$. Moreover, since the vertices in G are coloured independently, we have

$$\mathbb{P}(A_{e,c}) = \left(\frac{1}{8r}\right)^2 \tag{1.1}$$

Let $\mathcal{E}$ be the set of all events $A_{e,c}$ where an edge $e \in E$ has endpoints coloured $c \in F$ that is allowed to colour both endpoints of e. In order to show that the statement of this problem holds, we want to prove that $\mathbb{P}(\Omega \setminus \mathcal{E}) > 0$. For the dependency graph $D = (\mathcal{E}, \mathcal{F})$, a specific edge $e^{'} \in E$, and a specific colour $c \in F$ that is allowed to colour endpoints of the edge $e^{'}$, it is known from the Summatorial Local Lemma that this inequality automatically holds when

$$\sum_{\left((e,c),(e^{'},c^{'})\right) \in \mathcal{F}} \mathbb{P}(A_{e^{'},c^{'}}) \leq \frac{1}{4} \tag{1.2}$$

Since $\mathbb{P}(A_{e^{'},c^{'}})$ is known by the equation (1.1), we now want to find the number of incoming edges that the event $A_{e^{'},c^{'}}$ has in the dependency graph D. This can be done by finding the number of events in $\mathcal{E}$ on which the event $A_{e^{'},c^{'}}$ is dependent. Let $e^{'} = (i^{'}, j^{'})$ and $e = (i, j)$, it is obvious that the event $A_{e^{'},c^{'}}$ is mutually independent from the events $A_{e,c}$ where $i^{'} = i$ and $j^{'} = j$. That is, it remains to analyse the dependencies of the event $A_{e^{'},c^{'}}$ on the events $A_{e,c}$ where either $i^{'} = i$ or $j^{'} = j$. When $i^{'} = i$, it is obvious that the event $A_{e^{'},c^{'}}$ is dependent on the events $A_{e,c}$. That is, there must be an outgoing edge to the event $A_{e^{'},c^{'}}$ from each of these events in the dependency graph D, and thus the number of such edges is equal to the number of events $A_{e,c}$ when $i^{'} = i$. It is worth to note that the two endpoints of e in this case, which are $i^{'}$ and j, are coloured $c \in F_{i^{'}} \cap F_j \neq \emptyset$. For each colour $c \in F_{i^{'}}$, there are at most r neighbours j that $c \in F_j$, and thus there are at most r events $A_{e,c}$ for a specific colour $c \in F_{i^{'}}$. Moreover, since there are $8r$ colours in $F_{i^{'}}$, then the number of events $A_{e,c}$ in $\mathcal{E}$ when $i^{'} = i$ is at most $8r \times r = 8r^2$ in total. The same result can be obtained for the case where $j^{'} = j$ for similar reason. Therefore, the number of incoming edges that the event $A_{e^{'},c^{'}}$ has in the dependency graph D is at most

$16r^2$. Therefore, we have

$$\sum_{\big((e,c),(e',c')\big)\in\mathcal{F}} \mathbb{P}(A_{e',c'}) \leq \frac{16r^2}{64r^2} = \frac{1}{4}$$

this implies that the equation (1.2) holds, and the proof is finished. □

Problem 2. A hypergraph $H = (V, E)$ is a possible generalisation of a graph: Instead of $E \subseteq \binom{V}{2}$ consider a subset $E \subseteq 2^V$. We call the elements of E hyperedges.

An allowed 2-colouring of a hypergraph $H = (V, E)$ is a map $f : V \to [2]$, s.t. $\#f(e) = 2$ for all $e \in E$.

Let $H = (V, E)$ be a hypergraph, such that every hyperedge $e \in E$ connects at least k nodes (i.e. $\#e \geq k$ for every $e \in E$) and every hyperege $e \in E$ intersects at most d other hyperedges.

Show: If $e(d + 1) \leq 2^{k-1}$, then there is an allowed 2-colouring of H.

Proof.
Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space Ω is the set of all possible vertex colourings of the graph G using colours in $[2]$, and the probability measure $\mathbb{P}$ is given by independent colourings of vertices of G using colours in $[2]$ that are uniformly distributed. For any hyperedge $e \in E$, let A_e be the event in the probability space $(\Omega, \mathbb{P})$ where all vertices in e are coloured the same. For a specific colour $c \in [2]$, the probability that all vertices in e are coloured c is $\frac{1}{2^{|e|}}$, as there are 2 choices to colour a vertex in G and the colourings are independent. Therefore,

$$\mathbb{P}(A_e) = \frac{2}{2^{|e|}} \overset{|e|\geq k}{\leq} \frac{2}{2^k} = \frac{1}{2^{k-1}}$$

Let $\mathcal{E}$ be the set of all events A_e in the probability space $(\Omega, \mathbb{P})$ for any hyperedge $e \in E$. In order to show that the statement of the problem holds, we want to prove $\mathbb{P}(\Omega \setminus \mathcal{E}) > 0$. By the Symmetric Local Lemma, this inequality automatically holds when $ep(u + 1) \leq 1$ where p is the probability of any event in $\mathcal{E}$, and u is the upperbound on the number of events in $\mathcal{E}$ that any single event in $\mathcal{E}$ is dependent on. Consider a specific hyperedge $e \in E$, it is obvious that the event A_e is independent from the events $A_{e'} \in \mathcal{E}$ where the hyperedges $e, e^{'}$ do not intersect each other. Otherwise, the event A_e must be dependent on the events $A_{e'} \in \mathcal{E}$. Since the hyperedge e intersects with at most d hyperedges in G, then the maximum number of events $A_{e'} \in \mathcal{E}$, where e and $e^{'}$ intersect each other, that A_e is dependent on is d. We now want to verify whether the Symmetric Local Lemma holds, we have

$$ep(u + 1) = e\mathbb{P}(A_e)(d + 1) = \frac{e}{2^{k-1}}(d + 1)$$

Since $e(d + 1) \leq 2^{k-1}$, then $\frac{e}{2^{k-1}}(d + 1) \leq 1$. Therefore, the Symmetric Local Lemma holds, and the proof is finished. □

Problem 3. Let $G = (V, E)$ with $E \subseteq V \times V$ a directed graph. Let

$$\delta(G) = \min_{v\in V} \#\{w \in V \mid (v, w) \in E\}$$

the minimal out-degree of G and

$$\Delta(G) = \max_{v\in V} \#\{w \in V \mid (w, v) \in E\}$$

the maximal in-degree of G.

Show: If there is a $k > 1$, s.t $e \cdot (\Delta(G) \cdot \delta(G) + 1)(1 - 1/k)^{\delta(G)} < 1$, then there is a $i \geq 1$, such that G contains a simple cycle of length ik.

A simple cycle of length ℓ in a graph is a sequence $e_1, \ldots, e_\ell$ of edges, such that $e_i = (v_i, v_{i+1})$, $1 \leq i \leq \ell-1$ and $e_\ell = (v_\ell, v_1)$ and the vertices $v_1, \ldots, v_\ell$ are pairwise distinct.

Proof.
Since $e \cdot (\Delta(G) \cdot \delta(G) + 1)(1 - 1/k)^{\delta)G} < 1$, then $\delta(G) > 0$. Otherwise, the inequality is equivalent to $e < 1$, which is not true. As $\delta(G) > 0$, then the graph G must contain a simple cycle, this can be proved as follows. Consider the longest path $P = v_1, \ldots, v_\ell$ in the graph G. Since $\delta(G) > 0$, then v_ℓ must have an outgoing edge to another vertex in G as loops are not allowed in G. In case that v_ℓ has an outgoing edge to any vertex in G that is not in P, then there exists a path in G that is longer than P. However, this contradicts the fact that P is the longest path in G. Therefore, v_ℓ must have an outgoing edge to a vertex v_i in the path P where $1 \leq i \leq \ell - 1$, and thus the path $v_i, \ldots, v_\ell$ is a simple cycle in G.

Next, we want to determine whether G contains a simple cycle of length ik. In order to do this, we want to check all possible vertex colourings of G using colours in $[k]$ whether there exists a colouring such that out-neighbours of a vertex v is coloured $(c(v)+1) \pmod k$ where $c(v)$ is the colour of v. It is easy to verify that such a colouring exists when the graph G contains a simple cycle of length ik, and no vertex colouring of G satisfies this condition otherwise.

Instead of checking all vertex colourings of G, we want to do this in an arbitrary subgraph $G^{'}$ of G where each vertex in $G^{'}$ has exactly $\delta(G)$ outgoing edges. It is obvious that if $G^{'}$ contains a simple cycle of length ik, then G must also contain it. Based on this idea, let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space Ω is the set of all possible vertex colourings of the graph $G^{'}$ using the colours in $[k]$, which can be perceived as the set of all possible functions $f : V \to [k]$, and the probability measure $\mathbb{P}$ is given by independent colourings of vertices using colours in $[k]$ that are uniformly distributed.

For any vertex $v \in V$, let A_v be an event in the probability space $(\Omega, \mathbb{P})$ where all out-neighbours of v, which is denoted by $N^+(v)$, are not coloured by $(f(v) + 1) \pmod k$. For a specific vertex $v \in V$, the probability to not be coloured by a specific colour in $[k]$ is $(1 - 1/k)$. Moreover, as v has $\delta(G)$ out-neighbours, then

$$\mathbb{P}(A_v) = (1 - \frac{1}{k})^{\delta(G)}$$

Let $\mathcal{E}$ be the set of all events A_v in the probability space $(\Omega, \mathbb{P})$ for any vertex $v \in V$. In order to show that the statement of the problem holds, we want to prove $\mathbb{P}(\Omega \setminus \mathcal{E}) > 0$. By the Symmetric Local Lemma, this inequality automatically holds when $ep(d + 1) \leq 1$ where p is the probability of any event in $\mathcal{E}$, and d is the upperbound on the number of events in $\mathcal{E}$ that any single event in $\mathcal{E}$ is dependent on.

It is obvious that A_v is independent from A_w, where $v \neq w$, in case of $N^+(v) \cap \big(N^+(w) \cup w\big) = \emptyset$ since A_w only affects colouring of vertices which are not in $N^+(v)$, and thus the event A^w does not affect $\mathbb{P}(A_v)$. Otherwise, A_v must be dependent on A_w when $N^+(v) \cap \big(N^+(w) \cup w\big) \neq \emptyset$. That is, in order to determine the number of events A_w in $\mathcal{E}$ that any single vertex $v \in V$ is dependent on, for $v \neq w$, we consider the following cases:

- If $w \in N^+(v)$, then there are $\delta(G)$ events A_w in this case, as the vertex v has $\delta(G)$ out-neighbours.
- Otherwise, $w \in N^+(w) \cap N^+(v) \neq \emptyset$. It is known that each out-neighbour of the vertex v has at most $\Delta(G) - 1$ ingoing edges that are not from v. Moreover, the vertex v has $\delta(G)$ out neighbours. Therefore, there are at most $\delta(G) \cdot (\Delta(G) - 1)$ events A_w in this case.

Therefore, there are at most $\delta(G) + \delta(G) \cdot (\Delta(G) - 1) = \delta(G) \cdot \Delta(G)$ events A_w in $\mathcal{E}$ that any single vertex $v \in V$ that is dependent on. We now want to verify whether the Symmetric Local Lemma holds, we have

$$ep(d + 1) = e \cdot (\Delta(G) \cdot \delta(G) + 1)(1 - 1/k)^{\delta(G)}$$

Since $e \cdot (\Delta(G) \cdot \delta(G) + 1)(1 - 1/k)^{\delta(G)} < 1$, then the Symmetric Local Lemma holds, and the proof is finished. □